Foundations of Generative AI (ICT4403)
Assignment 2 Submission

Submission Details & GitHub Repository Access Link

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Assignment Title	Foundations of Generative AI (ICT4403) - Assignment 2
System Title	Autoregressive Text Generation System using Simple RNN, Bidirectional RNN, LSTM, and GRU
Dataset Name & Link	The Adventures of Sherlock Holmes by Arthur Conan Doyle URL: https://www.gutenberg.org/files/1661/1661-0.txt
GitHub Repository Link	https://github.com/AkCodes23/FGAI_Assignment.git

System Overview & Implementation Summary

- Models Implemented: Simple RNN (Elman), Bidirectional RNN, LSTM, and GRU under matched capacity (embedding dim: 128, hidden dim: 256).
- Evaluation Measures: Cross-Entropy Loss, Perplexity (PPL), Top-1 Accuracy, Top-5 Accuracy, Training Time, and Inference Latency.
- Qualitative Evaluation: Coherence, Lexical Diversity (Distinct-1 & Distinct-2), and 3-Gram Repetition Rate across sampling temperatures T = 0.2, 0.7, 1.2.
- Full Reproducibility: Code, checkpoints, logs, plots, tests, and Jupyter Notebook (.ipynb) are organized in the GitHub repository.